CIVICSIFT AI

WEEK 6 — FINAL PROJECT REPORT

An Evidence-Aware NLP Framework for Citizen Complaint Analytics

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Project domain: Natural Language Processing

Reporting period: Weeks 1–6

Submission: Final integrated report

November 2026

Report Scope and Evidence Statement

This report consolidates the five weekly CivicSift AI submissions into one final account of the project's planning, methodology, implementation, measured findings, and remaining work. It is written to distinguish completed experiments from proposed designs. The words "implemented", "proposed", and "illustrative" are used deliberately.

Important accuracy note Week 1 is a proposal; Week 2 is a data-preparation methodology plan; Week 3 contains an implemented classification experiment; Week 4 is an NER design; and Week 5 contains an implemented topic-modelling and summarisation experiment. The report does not represent the unimplemented parts as completed software.

Project status at a glance

Workstream	Status evidenced in weekly submissions	Evidence level
Project planning (Week 1)	Problem definition, objectives, research direction, tools, risks and evaluation plan documented.	Planning
Data preparation (Week 2)	Collection, governance, annotation, preprocessing and feature-engineering methodology documented; no completed data-preparation claim.	Methodology
Text classification (Week 3)	TF-IDF pipelines and three classical classifiers implemented and evaluated on a held-out split.	Implemented experiment
NER (Week 4)	Entity schema, multilingual pipeline, BIO tagging, model strategy and evaluation design documented.	Design only
Topic modelling & summaries (Week 5)	Preprocessing, LDA, TF-IDF centroid summaries and evaluation scripts executed.	Implemented experiment

The experiments use related NYC 311 complaint data but not necessarily an identical frozen snapshot or identical fields. Week 3 and Week 5 counts are therefore reported separately.

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1. Executive Summary

CivicSift AI is a proposed evidence-aware Natural Language Processing framework intended to help authorised reviewers organise citizen complaints about public services. The concept responds to a practical challenge: complaint messages are often short, informal, repetitive and difficult to sort consistently. The planned system brings together complaint-category suggestions, urgency cues, entity extraction, semantic similarity and topic-level summaries, while keeping a human reviewer responsible for decisions.

Across six weeks, the work progressed from a project proposal to two implemented NLP experiments and two detailed methodology/design documents. Week 3 tested TF-IDF text classification with Multinomial Naive Bayes, Logistic Regression and Linear Support Vector Classification. On the reported 51-record test split, Linear SVM achieved the strongest result: 49.02% accuracy and 47.13% weighted F1. This is a baseline result, not evidence of production-grade classification.

Week 5 processed 369 complaint records, retaining 366 cleaned records. An LDA model generated four topics. Evaluation reported perplexity 293.686213, topic diversity 1.000000, average topic-assignment confidence 0.611134 and average summary relevance 0.318100. The outputs show that the pipeline runs end to end, while mixed topics and moderate summary relevance indicate room for improvement.

The NER work remains a design rather than a trained model. The supplied NYC 311 data contains short phrases and record-level categories, not token-level entity spans or verified multilingual coverage. As a result, no NER precision, recall or F1 is claimed. Likewise, Week 2 documents a proposed preparation method and does not claim that its broader multilingual dataset was collected or annotated.

The main outcome is a documented, reproducible prototype foundation: an implemented English-language baseline for classification and topic analytics, plus a considered roadmap for multilingual data governance, NER, urgency assessment, duplicate discovery and human-reviewed deployment.

Key outcomes

- A clearly scoped citizen-complaint analytics problem and human-in-the-loop concept.
- A documented privacy-aware data collection and preprocessing strategy.
- An implemented TF-IDF classification comparison with a reported best baseline.
- A proposed NER schema and evaluation plan, with dataset gaps explicitly identified.
- An implemented LDA and extractive summarisation pipeline with quantitative evaluation outputs.
- A consolidated roadmap that separates demonstrated capabilities from future work.

2. Introduction and Civic Problem

Public-service organisations may receive complaints about roads, water supply, sanitation, drainage, street lighting, noise, construction and public safety. Such information can arrive as free-text descriptions rather than standardised forms. Different people may describe the same issue using different words, spelling conventions or languages. Messages may also be incomplete or duplicated.

Manual triage can be time-consuming and may produce inconsistent categories. Keyword matching can miss semantically related complaints when they use different vocabulary. At the same time, automated systems can make mistakes, particularly when training data is small, imbalanced or not representative of the languages used by residents.

CivicSift AI frames NLP as a support tool rather than an autonomous authority. Its intended role is to organise information, suggest categories, surface possible urgency indicators and retrieve related complaints for review. The system should show evidence and uncertainty, preserve the original message and allow authorised personnel to correct its output.

2.1 Why an evidence-aware approach matter

- A category prediction should be presented as a suggestion, not a confirmed administrative decision.
- An urgency cue should be traceable to words or phrases in the complaint; sentiment alone is not a reliable proxy for risk.
- A similarity score identifies candidate related reports, not proof that two reports describe the same incident.
- Extracted locations are text mentions unless independently verified; they should not be treated as confirmed geographic facts.
- Low-confidence or conflicting outputs should be referred to human review.

2.2 Intended users and boundary

The primary intended users are authorised complaint reviewers, analysts and public-service teams. CivicSift AI is not designed to replace official complaint channels, determine legal liability, make emergency-response decisions, or automatically close or merge citizen reports.

3. Project Aim, Objectives and Scope

3.1 Aim

To design and evaluate a lightweight, human-assisted NLP framework that can organise citizen complaints, extract useful text information and support incident discovery.

3.2 Objectives

1. Define a complaint taxonomy for public-service issues such as water, roads, sanitation, drainage, electricity and safety hazards.
2. Document a privacy-conscious data strategy and reproducible preprocessing approach.
3. Build a TF-IDF and classical machine-learning baseline for complaint-category classification.
4. Design a multilingual NER pipeline for standard and civic-specific entities.
5. Apply topic modelling to discover recurring patterns in complaint text.
6. Generate representative extractive summaries to make topic outputs easier to inspect.
7. Evaluate implemented components with suitable metrics and record limitations.
8. Recommend a responsible path towards a human-reviewed prototype.

3.3 Scope actually demonstrated

Included in the documented project	Not demonstrated as completed
NYC 311-based English complaint-text experiments.	A deployed public-facing application or production integration.
TF-IDF classification using three classical models.	A validated multilingual classifier for Hindi, Kannada and code-mixed messages.
LDA topic modelling and TF-IDF-based extractive summaries.	A trained/evaluated NER model.
Evaluation outputs for the implemented Week 3 and Week 5 experiments.	A trained urgency classifier or validated urgency labels.
Proposed privacy, annotation and human-review controls.	A validated semantic duplicate-detection module.

3.4 Assumptions

The project assumes that suitable public or consented data can be used under applicable terms, that complaint categories can be defined consistently, and that human reviewers can inspect uncertain results. These assumptions require verification before deployment.

4. Background and Related Work

The weekly reports identify several relevant NLP directions. TF-IDF provides a transparent and computationally efficient representation for sparse text classification. Logistic Regression and linear SVM models are common baselines for text represented by TF-IDF features. LDA provides a probabilistic approach to discovering latent topic structure in a collection of documents. Sentence embeddings can support semantic retrieval, while transformer token-classification models can be adapted for NER.

Method / resource	Role considered in CivicSift AI	Evidence status
TF-IDF	Word and n-gram features for classification and summarisation.	Implemented in Weeks 3 and 5.
Logistic Regression	Classical category-classification baseline.	Implemented in Week 3.
Linear SVM	Linear classifier for sparse TF-IDF features.	Implemented in Week 3; strongest reported result.
Multinomial Naive Bayes	Probabilistic text-classification baseline.	Implemented in Week 3.
LDA	Discover recurring themes across complaint text.	Implemented in Week 5.
mBERT / XLM-R / IndicBERT	Candidate multilingual transformer families for NER or classification.	Proposed candidates only.
Sentence-BERT / sentence embeddings	Candidate approach for semantic duplicate discovery.	Proposed in Week 1/2; not evaluated in reported weeks.

The proposal cites IndicNLPSuite and IndicBERT resources as motivation for future Indian-language coverage. It also cites research on multilingual representations and Hindi-English code-mixed NLP. These references motivate evaluation; they do not establish that CivicSift AI itself has multilingual performance.

4.1 Research opportunity

The project's intended contribution is the combination of complaint classification, evidence-aware urgency suggestions, entity extraction and semantic duplicate discovery in a human-in-the-loop workflow. The proposal does not claim that multilingual complaint classification is entirely new. The contribution is framed as a practical integration and evaluation opportunity.

5. Dataset and Source Context

The project's implemented experiments use a small NYC 311 complaint dataset derived from the NYC Open Data 311 Service Requests resource. The Week 3 report describes retrieving data through the Socrata CSV endpoint and using complaint_type and descriptor. The Week 4 NER report describes a supplied file named nyc_311_complaints.csv with text and complaint_type columns. The Week 5 implementation uses a complaint-text field and reports its own preprocessing counts.

The NYC dataset provides a relevant public-service domain for a proof-of-concept because it contains complaint descriptions and service categories. However, it is not a representative sample of complaints from Indian cities and does not establish Hindi, Kannada or code-mixed language coverage.

Weekly experiment / design	Dataset facts reported	Use in this final report
Week 3 classification	369 fetched; 203 usable after dropping missing required values; 24 complaint categories; 152 train / 51 test.	Classification results only.
Week 4 NER assessment	369 rows; 2 columns (text, complaint_type); 0 missing; 348 unique texts; 107 unique complaint types.	NER suitability assessment only.
Week 5 topic modelling	369 input records; 366 retained after cleaning; 3 removed; 0 empty cleaned texts.	Topic and summary results only.

Do not merge these counts The weekly reports describe different processing stages, selected fields and/or data snapshots. This report preserves each reported count in its own experimental context rather than forcing them into one common total.

5.1 Data source and reproducibility

Week 3 records the endpoint https://data.cityofnewyork.us/resource/erm2-nwe9.csv. The broader source is NYC Open Data's 311 Service Requests dataset. A future reproducible release should record the retrieval date, query parameters, selected columns, data version, licence/terms, and a cryptographic checksum for the exact local file.

5.2 Data suitability

- Useful for civic vocabulary and initial category experiments.
- Short descriptors may contain less context than full complaint narratives.
- Category frequencies may be imbalanced.
- The data does not contain token-level entity annotations required for supervised NER.
- Language metadata is not reported as part of the supplied file; multilingual coverage cannot be inferred.

6. Week 1 — Planning and Proposal

Week 1 established the project title: "CivicSift AI – An Evidence-Aware Multilingual NLP Framework for Citizen Complaint Triage and Incident Discovery." The proposal described unstructured citizen complaints as the main input and positioned NLP as a tool for first-pass organisation.

6.1 Planned functions

- Complaint category suggestion for common public-service issues.
- Urgency estimation using labelled examples and transparent text cues.
- Information extraction for locations, dates, services and infrastructure mentions.
- Semantic similarity retrieval to surface potentially related reports.
- A simple interface showing predictions, confidence, evidence phrases and related complaints.
- Human review and correction before a record is treated as confirmed.

6.2 Proposed language scope

The initial target languages were English, Hindi and Kannada, with controlled evaluation of Romanised and code-mixed text. The proposal recognised that local-language coverage would require representative examples, appropriate preprocessing and language-specific evaluation.

6.3 Planned methods and tools

Area	Proposed tools / approach
Programming and data	Python, Pandas, NumPy
Text processing	Regular expressions, NLTK or spaCy; language-specific resources where suitable
Classification	TF-IDF with Logistic Regression or Linear SVM; later comparison with IndicBERT or XLM-R
Urgency	Transparent rules as baseline; supervised model only if enough labels exist
Duplicate retrieval	Multilingual sentence embeddings and cosine similarity
Interface	Streamlit
Evaluation	Macro-F1, per-class precision/recall, confusion matrix, retrieval metrics, language-wise error analysis

6.4 Proposed targets versus achieved results

The proposal listed provisional targets such as category-classification macro-F1 of at least 0.75, critical-class urgency recall of 0.80 where data permits, and duplicate retrieval Precision@5 of 0.75. These were planning targets only. They are not achieved results and should not be reported as such.

6.5 Planning risks

Week 1 identified risks including limited labelled data, language imbalance, code-mixed spelling variation, false urgency escalation, false duplicate matches, privacy exposure and compute constraints. Mitigations included pilot annotation, data minimisation, human review, confidence thresholds and transparent reporting.

7. Week 2 — Data Collection and Preparation Methodology

Week 2 translated the proposal into a detailed methodology for data collection, governance, annotation, preprocessing and feature engineering. The report explicitly states that it is a methodology and preparation plan: it does not claim that a dataset was downloaded, annotated, cleaned or evaluated as part of Week 2.

7.1 Proposed collection strategy

- Use public datasets only after checking provenance, licence and permitted use.
- Use consented examples or clearly labelled synthetic data where appropriate.
- Record source, language, script, category, urgency label and duplicate-group identifier where available.
- Retain provenance and transformation history for reproducibility.
- Avoid unnecessary personal data and mask identifiers where required.

7.2 Proposed record schema

The planned record structure includes complaint text, language/script metadata, labels, source provenance and duplicate-group information. Optional location mentions are treated as text-derived information rather than verified location facts.

7.3 Preprocessing principles

- Preserve the original complaint text for audit and reviewer comparison.
- Apply Unicode normalisation and conservative whitespace/markup cleanup.
- Preserve negation, numbers, dates and urgency cues because they may affect meaning.
- Use language-appropriate tokenisation and avoid forcing code-mixed messages into a single-language representation.
- Handle transliteration and spelling variation conservatively; document transformations.
- Avoid indiscriminate stop-word removal when it could remove contextual meaning.

7.4 Feature engineering choices

Feature family	Intended task	Rationale in Week 2
Word-level TF-IDF	Category classification	Interpretable sparse baseline for recurring terms and phrases.
Character n-gram TF-IDF	Spelling variants and informal text	Can capture subword patterns and variation.
Multilingual sentence embeddings	Duplicate/related-report retrieval	Can compare semantic meaning beyond exact word overlap.
IndicBERT / XLM-R	Potential advanced classification	Candidate contextual representations to evaluate later.

7.5 Quality assurance and governance

The methodology recommends checking missing values, duplicates, category balance, language/script coverage and annotation consistency. It also recommends a source register, annotation guide,

reproducible preprocessing, versioned pilot dataset and group-aware data splits. The report's explicit status note is retained here: these are planned controls, not verified completed Week 2 outputs.

8. Week 3 — Text Classification Implementation

Week 3 implemented and compared three classical text classifiers using TF-IDF features. The selected input field was descriptor and the target was complaint_type. The experiment used the NYC Open Data Socrata CSV endpoint recorded in the weekly report.

8.1 Dataset preparation

Stage	Reported value
Rows fetched	369
Rows retained after removing missing required values	203
Target classes	24 complaint categories
Training set	152 records
Test set	51 records
Split strategy	Stratified 75:25 train-test split

The dataset size is small relative to the number of classes. This makes the result sensitive to the distribution of examples and means that per-class performance may be unstable.

8.2 Feature extraction and leakage control

The implementation used TF-IDF with English stop-word removal and unigrams plus bigrams (ngram_range=(1,2)). The vectorizer was placed inside the scikit-learn pipeline so it was fitted on training data rather than on the full dataset. This reduces the risk of information leakage from the test split.

8.3 Models compared

- Multinomial Naive Bayes — probabilistic baseline suitable for count-like text representations.
- Logistic Regression — linear discriminative classifier over sparse TF-IDF features.
- Linear Support Vector Classifier — margin-based linear classifier commonly used with sparse text features.

8.4 Evaluation

The report compared accuracy, weighted precision, weighted recall and weighted F1. The best model on the reported held-out split was Linear SVM.

Model	Reported relative outcome
Linear SVM	Best: 49.02% accuracy; 47.13% weighted F1.
Logistic Regression	Second-best among the three models.
Multinomial Naive Bayes	Lowest-performing of the three models.

The report does not provide all individual metric values for every model in the consolidated evidence available here; therefore, only the specific best-model values and relative ranking are stated.

8.5 Interpretation

A weighted F1 of 47.13% indicates that the model provides a starting point but still makes substantial classification errors. The result should not be interpreted as evidence that complaint triage can be automated reliably. A larger, more representative dataset, per-class error analysis and cross-validation would be needed.

9. Week 4 — Named Entity Recognition Pipeline Design

Week 4 proposed an end-to-end NER design for extracting useful spans from citizen complaints. The proposed labels include standard entities PERSON, LOCATION, ORGANIZATION, DATE and TIME, alongside CivicSift-specific ISSUE and INFRASTRUCTURE labels.

9.1 Entity schema

Label	Meaning	Illustrative example from design
PERSON	Name of a person	Rahul
LOCATION	Place, street, area, city or landmark	MG Road; Bengaluru
ORGANIZATION	Government body, department, company or institution	Municipal Corporation
DATE	Calendar date or date expression	10 November
TIME	Time expression	8:30 PM
ISSUE	Civic problem being reported	water leakage; pothole
INFRASTRUCTURE	Physical public asset or facility	road; hydrant; streetlight

9.2 Proposed processing pipeline

1. Preserve the original complaint and identify language/script where feasible.
2. Apply conservative Unicode and text normalisation while retaining punctuation, numbers and token offsets.
3. Tokenise using a language-appropriate method and align labels with model subwords.
4. Run a multilingual token-classification model candidate such as mBERT, XLM-R or an Indic-focused model.
5. Apply BIO sequence checks, merge subword spans and normalise date/time outputs without losing original text.
6. Use confidence thresholds and send uncertain or conflicting spans to human review.
7. Return structured entities with character offsets, confidence, source complaint ID and model/version metadata.

9.3 Data gap and status

NER status The Week 4 document is a design and implementation strategy. It does not claim that an NER model was trained, deployed or evaluated.

The inspected NYC file contained 369 rows, two columns (text and complaint_type), zero missing values, 348 unique complaint texts and 107 unique complaint types. It contains short phrases and record-level categories, not token-level NER spans. The complaint_type field cannot be used as PERSON, LOCATION, DATE or other token-level entity annotation.

The file also does not establish multilingual coverage. Hindi, Kannada and code-mixed support remain project requirements, not demonstrated dataset properties. Week 4 recommends investigating a suitable public multilingual NER resource such as Naamapadam after checking the exact release, languages, schema and licence, and manually annotating civic ISSUE and INFRASTRUCTURE examples.

9.4 Evaluation plan

The proposed evaluation is exact-span entity-level precision, recall and F1, reported by entity label and language. False positives, false negatives, boundary errors and confusion between ISSUE and INFRASTRUCTURE should be inspected. No NER metric is reported because no annotated held-out test set was evaluated.

10. Week 5 — Topic Modelling and Extractive Summarisation

Week 5 implemented an NLP pipeline using NYC 311 complaint descriptions. The work covered preprocessing, Latent Dirichlet Allocation (LDA), TF-IDF-based extractive summarisation and evaluation.

10.1 Preprocessing

Measure	Reported result
Input records	369
Processed records	366
Removed records	3
Empty cleaned texts	0

The script lowercased text, applied regular-expression cleaning, filled missing text and removed English stop words using scikit-learn's ENGLISH_STOP_WORDS. The cleaned output was saved as preprocessed_complaints.csv.

10.2 LDA configuration

The implementation used CountVectorizer with max_features=1000, min_df=2, max_df=0.95 and ngram_range=(1,2). The resulting document-term matrix had shape (366, 195), with a vocabulary of 195 terms. An LDA model was configured with four topics.

Topic	Count	Share	Keywords recorded
1	155	42.35%	blocked, sidewalk, defective, permit, illegal, work, sign, improper, window, garbage
2	66	18.03%	loud, music, truck, construction, door, music party, loud music, party, loud talking, talking
3	67	18.31%	street, missing, street light, light, basin, store, catch basin, catch, line, trash
4	78	21.31%	food, signal, hydrant, odor, veh, non, building, passenger, driver, veh signal

10.3 Extractive summarisation

The summarisation stage used TF-IDF representations, a topic centroid and cosine similarity to select representative complaint descriptions. The implementation produced topic_summaries.csv and topic_summaries.txt. This is extractive selection: it selects representative existing descriptions rather than generating new sentences.

10.4 Representative descriptions

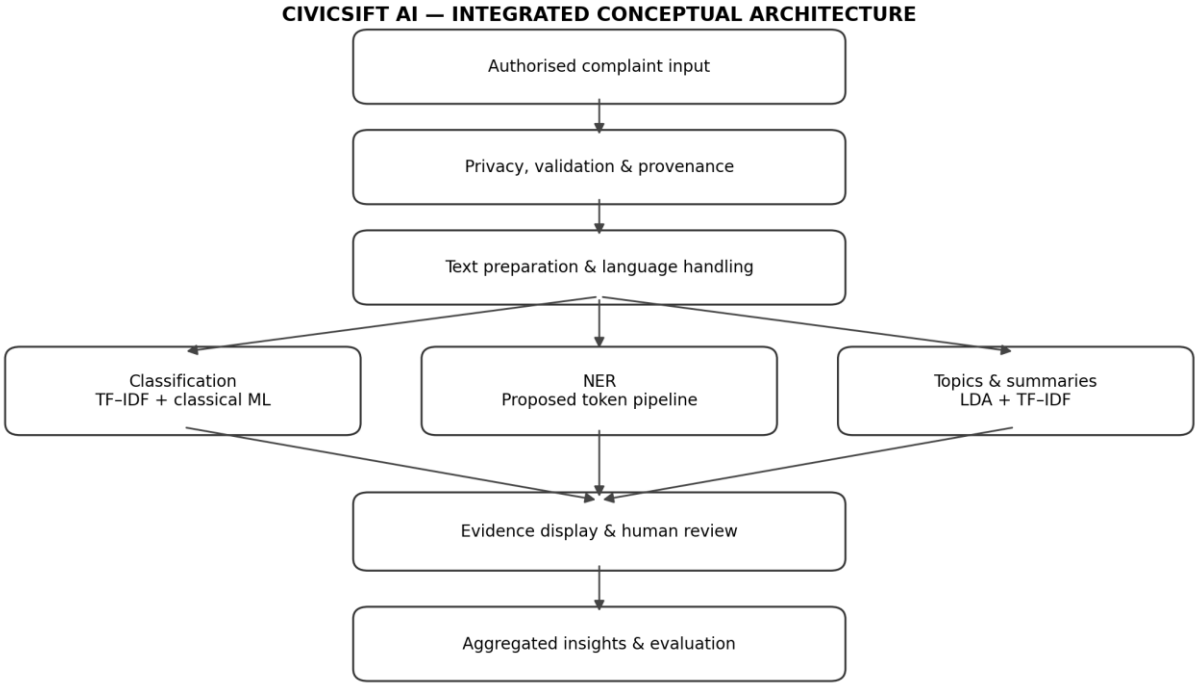
Topic	Representative descriptions recorded
1	Blocked Sidewalk; Broken Sidewalk; Garbage or Litter; Traffic Sign or Signal Blocked; Sign/Awning/Marquee - Illegal/No Permit.
2	Lamppost Damaged; No Water; DOOR; DOOR FRAME; DOOR/FRAME.
3	Street Light Out; Trash or Recycling; Blocking Street; Clear Street Light; Street Light

	Dayburning.
4	Non-Food Vendor; Veh Signal Lamp; Fire Hydrant; Food Vendor; Fire Hydrant Running.

The recorded category examples include Illegal Parking, General Construction/Plumbing and Sidewalk Condition for Topic 1; Noise, Water Maintenance and Street Light Condition for Topic 2; Consumer Complaint, Street Light Condition and Street Condition for Topic 3; and Traffic Signal Condition, Food Establishment and Water Maintenance for Topic 4.

11. Integrated Workflow and Architecture

The integrated view connects the project's implemented components with its proposed components. It should be read as a conceptual workflow and not as evidence that all modules are already connected in one deployed application.



11.1 Workflow stages

Stage	Input	Operation	Output / control
1. Intake	Authorised complaint text	Validate input; assign record ID; retain provenance.	Raw complaint retained.
2. Privacy preparation	Raw text	Minimise or mask unnecessary identifiers; preserve original.	Prepared text + audit trail.
3. Language and text handling	Prepared text	Detect language/script where feasible; normalise carefully.	Language metadata; token-safe text.
4. Classification	Text features	TF-IDF and classical model prediction.	Suggested complaint category; confidence where available.
5. NER (future)	Tokenised complaint	Multilingual token classification and post-processing.	Entity spans; confidence; review flag.
6. Topic analytics	Complaint corpus	LDA topic assignments and representative extraction.	Topic distribution and summaries.
7. Human review	Predictions and	Reviewer confirms,	Reviewed record.

	evidence	corrects or rejects suggestions.	
8. Monitoring	Reviewed records and test sets	Metrics, error analysis and data-quality checks.	Evaluation and improvement backlog.

11.2 Component status

Classification and topic analytics were implemented in separate weekly experiments. NER remains a design. The Week 1 urgency and duplicate-retrieval modules remain proposed directions. The integrated architecture is therefore a target architecture that shows how the pieces could fit together after additional implementation and validation.

12. Results Consolidation and Interpretation

12.1 Classification results

Measure	Result / finding	Interpretation
Best model	Linear SVM	Strongest of the three compared models in this experiment.
Accuracy	49.02%	Correct predictions on fewer than half of the 51 held-out examples.
Weighted F1	47.13%	Moderate baseline; class-frequency weighting does not remove minority-class concerns.
Dataset size	203 usable records; 24 categories	Small multi-class sample limits generalisation.

12.2 Topic modelling and summary results

Metric	Reported value	Interpretation / caution
LDA perplexity	293.686213	A model-specific measure; compare only under matching preprocessing/vectorisation settings.
Topic diversity	1.000000	A code-defined diversity score; it does not establish perfect semantic coherence.
Average topic assignment confidence	0.611134	Average assignment confidence as defined by the implementation; not a universal quality score.
Average summary relevance	0.318100	Indicates limited-to-moderate similarity under the implemented relevance calculation.

Topic	Summary relevance
Topic 1	0.2660
Topic 2	0.3278
Topic 3	0.3468
Topic 4	0.3318

12.3 Combined interpretation

- The project demonstrates that conventional NLP methods can run on a small civic-complaint dataset and produce measurable outputs.
- Classification performance is limited, so the model is best treated as a baseline for experimentation.
- LDA produced four distinguishable keyword groups, but some topics contain mixed or loosely related terms.
- Extractive summaries provide examples for manual inspection, but relevance scores indicate that summary quality can be improved.
- No quantitative result is available for NER, urgency estimation, multilingual robustness or duplicate retrieval.

13. Evaluation Strategy and Quality Assurance

13.1 Evaluation used

Week 3 used accuracy, weighted precision, weighted recall and weighted F1 for category classification. Week 5 reported LDA perplexity, a code-defined topic-diversity score, average topic-assignment confidence and TF-IDF/cosine-based summary relevance.

13.2 Evaluation not yet completed

Component	Recommended evaluation before claiming readiness
Classification	Stratified repeated cross-validation; macro-F1; per-class precision/recall/F1; confusion matrix; error review.
NER	Exact-span entity precision, recall and F1 overall, by entity label and language.
Urgency	Recall for critical examples, false-negative review, calibration and evidence-based human review.
Duplicate retrieval	Precision@k, Recall@k, pairwise F1 or cluster purity using reviewed pairs.
Topic modelling	Manual topic coherence review, stability across seeds, topic count sensitivity and domain expert inspection.
Summarisation	Human relevance assessment and ROUGE only if reliable reference summaries are created.
Application	Functional, privacy, security, usability and latency tests.

13.3 Why ROUGE was not reported

The Week 5 experiment did not have reference summaries against which generated or extracted summaries could be compared. Therefore, ROUGE was not calculated. The implemented summary-relevance measure was used instead, and its definition should be retained with the code for reproducibility.

13.4 Reproducibility checklist

- Record exact dataset snapshot and retrieval date.
- Record preprocessing code and library versions.
- Keep train, validation and test splits separate.
- Fit vectorisers on training data only for supervised classification.
- Set and record random seeds where applicable.
- Save model artifacts and evaluation outputs.
- Report class distribution and per-class errors.
- Keep illustrative examples separate from actual dataset records.

14. Privacy, Ethics and Human Oversight

Citizen complaints may contain personal names, contact details, precise locations or other sensitive information. The Week 1 and Week 2 reports therefore recommend data minimisation, masking unnecessary identifiers, provenance tracking and controlled access. Only data with appropriate permission should be used. Synthetic examples should be labelled as synthetic and should not be presented as real reports or measured predictions.

14.1 Responsible data handling

- Collect only fields required for the stated analytics purpose.
- Check source terms and licences before collection or redistribution.
- Preserve original text in a controlled location for audit, while minimising exposure in model outputs.
- Apply access controls and retention/deletion rules appropriate to the deployment context.
- Document preprocessing and transformations.
- Do not expose personal information in dashboards, screenshots or public repositories.

14.2 Human-in-the-loop controls

A reviewer should be able to inspect the source complaint, model suggestion, evidence phrases, confidence and any extracted spans. Low-confidence or conflicting results should be flagged. The system should not automatically escalate, merge or close complaints without authorised review.

14.3 Fairness and language representation

If a future version supports English, Hindi, Kannada, transliteration and code-mixed messages, each language/script should be represented in training and evaluation. Overall metrics alone may conceal poor performance for a smaller language group. Performance gaps should be reported, and unsupported language inputs should be flagged rather than silently processed as if supported.

15. Challenges and Limitations

15.1 Dataset limitations

- The implemented experiments use small data samples relative to the number of categories.
- Week 3 uses short descriptors rather than complete complaint narratives.
- Class imbalance may affect both training and weighted metrics.
- NYC 311 data is geographically and institutionally specific; it does not prove transfer to Indian civic contexts.
- The supplied dataset does not provide verified multilingual coverage or token-level NER labels.

15.2 Model and method limitations

- The Week 3 held-out test set contains only 51 records, so results may vary with a different split.
- No extensive hyperparameter search or cross-validation is reported in the Week 3 experiment.
- The best model's 49.02% accuracy and 47.13% weighted F1 are not sufficient to justify autonomous triage.
- Four LDA topics were selected as an initial experiment; the reports do not establish that four is optimal.
- Topic 2 and Topic 4 include mixed terms and may not represent clean semantic themes.
- Topic diversity of 1.0 is the code-defined metric, not proof of ideal topics.
- Summary relevance is modest and there are no reference summaries for ROUGE.
- NER has no trained model or measured metrics.
- Urgency estimation and semantic duplicate retrieval were planned but not implemented in the reported experiments.

15.3 Integration limitations

The weekly experiments are separate scripts and workflows. The final architecture is a conceptual integration. The reports do not establish a single connected application, a deployed dashboard, production security controls or operational performance.

15.4 Evidence boundary

Interpretation rule A planned capability is not a measured capability. This final report retains that distinction so that project achievements remain credible and verifiable.

16. Recommendations and Future Scope

16.1 Data improvements

9. Create a versioned, documented dataset with full complaint narratives where permitted.
10. Expand the sample size and improve representation across categories, locations and languages.
11. Create a consistent annotation guide for category, urgency, entity spans and duplicate groups.
12. Use incident-group-aware splits to prevent near-duplicate leakage.
13. Collect or annotate representative Hindi, Kannada, Romanised and code-mixed examples before claiming multilingual support.

16.2 Model improvements

14. Tune TF-IDF settings and compare word-level with character-level features.
15. Use repeated stratified cross-validation and report macro-F1 and per-class results.
16. Investigate calibrated probabilities or confidence thresholds and an abstention option.
17. Evaluate transformer candidates only after sufficient labelled data is available.
18. For NER, annotate spans and compare mBERT, XLM-R and Indic-focused models on the same held-out set.
19. For duplicate discovery, evaluate sentence embeddings with reviewed positive/negative pairs.
20. For topic modelling, test different topic counts and seeds, and manually review topic coherence.

16.3 Product and governance improvements

21. Build a small reviewer-facing interface only after model outputs and data handling are stable.
22. Show source text, predicted label, evidence and uncertainty together.
23. Add role-based access, logging, retention controls and secure storage before using real citizen data.
24. Keep human approval for urgency escalation, duplicate merging and final categorisation.
25. Monitor model errors and language-wise performance after deployment.

17. Potential Applications

If future development validates the proposed components, CivicSift AI could support several public-service workflows. These are potential applications rather than deployments demonstrated by the six-week project.

Potential application	Possible use	Required safeguard
Complaint triage	Suggest a first-pass service category for reviewer confirmation.	Human confirmation; monitor per-class errors.
Issue trend monitoring	Summarise recurring complaint themes over a time period.	Check data coverage and topic stability.
Related complaint discovery	Retrieve candidate reports that may concern the same incident.	Reviewer confirmation before merging.
Information extraction	Highlight location, date, organisation, issue and infrastructure mentions.	Treat extracted text as unverified until checked.
Operational dashboards	Display category counts and topic summaries for analysts.	Privacy controls and aggregation thresholds.
Multilingual access	Support complaints in additional Indian languages.	Language-specific data, evaluation and uncertainty flags.

17.1 Example use scenario (illustrative)

A resident submits a short message about a damaged streetlight near a named road. In a future system, a classifier might suggest a street-light category, an NER component might highlight the road name and infrastructure phrase, and semantic retrieval might show similar reviewed reports. The reviewer would inspect the original message and decide whether the category and related-report suggestions are appropriate.

Example status This scenario is illustrative. It is not a prediction produced by the Week 3 or Week 5 experiments and is not a record from the supplied dataset.

18. Conclusion

CivicSift AI progressed from a project concept into a documented set of NLP methods and two implemented experiments. The project defined a meaningful civic complaint analytics problem, developed a privacy-aware data-preparation methodology, compared classical text classifiers, designed a multilingual NER pipeline and implemented topic modelling with extractive summaries.

The strongest measured classification baseline was Linear SVM, with 49.02% accuracy and 47.13% weighted F1 on the reported held-out split. The Week 5 pipeline retained 366 of 369 records, produced four LDA topics and generated representative complaint descriptions. Its reported evaluation values provide a baseline for further experimentation, not proof of semantic perfection.

The most important limitation is that the integrated multilingual system remains a proposed architecture rather than a fully implemented and deployed application. Week 2 is a methodology plan, Week 4 is NER design only, and urgency estimation and semantic duplicate retrieval were not measured in the reported implementation. The available NYC data also does not establish Indian-language coverage or token-level NER annotations.

The next stage should focus on larger and more representative datasets, consistent annotation, stronger validation, language-wise evaluation and careful human-reviewed integration. By preserving evidence, uncertainty and accountability, CivicSift AI can develop responsibly from a student NLP prototype into a more useful decision-support tool.

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The above references are consolidated from the weekly reports. Dataset/model release details and licences should be rechecked before any new data collection or deployment.

Appendix A — Implementation Artefacts and Outputs

A.1 Week 3 artefacts

- 01_data_preparation.py — prepares the classification dataset.
- 02_train_models.py — trains and compares the three classifiers.
- 03_predict_complaint.py — accepts a complaint and returns a predicted category.
- Prepared CSV, saved model artefacts and model comparison results, as recorded in the weekly implementation.

A.2 Week 5 artefacts

- 01_preprocessing.py — cleans complaint text and saves preprocessed_complaints.csv.
- 02_topic_modeling.py — fits LDA and saves topic outputs and model/vectorizer artefacts.
- 03_text_summarization.py — selects representative descriptions using TF-IDF and cosine similarity.
- 04_evaluation.py — calculates evaluation metrics and saves evaluation outputs.

A.3 Week 5 output files

Output	Purpose
preprocessed_complaints.csv	Cleaned complaint text.
lda_topics.csv	Topic keyword information.
complaints_with_topics.csv	Complaint records with assigned topics.
lda_model.pkl / count_vectorizer.pkl	Saved LDA and vectorizer artefacts.
topic_summaries.csv / topic_summaries.txt	Representative complaint descriptions by topic.
evaluation_metrics.csv	Aggregate evaluation metrics.
summary_relevance_by_topic.csv	Topic-wise summary relevance values.

Appendix B — Metric Definitions

Metric	Definition / use	Important interpretation
Accuracy	Correct predictions divided by all predictions.	Can conceal poor minority-class performance.
Precision	True positives divided by predicted positives.	Measures reliability of positive predictions for a class.
Recall	True positives divided by actual positives.	Measures how many actual class instances are found.
F1	Harmonic mean of precision and recall.	Balances precision and recall.
Weighted F1	Class F1 values weighted by class support.	Reflects class frequency; does not give equal weight to every class.
LDA perplexity	A measure related to how well a probabilistic topic model predicts held-out text.	Compare only under consistent data and vectorisation settings.
Topic diversity	In this project, a code-defined measure of unique topic keywords.	Not equivalent to semantic coherence.
Cosine similarity	Similarity between vector representations based on angle.	A relevance proxy, not a human quality judgement.
Exact-span NER F1	Entity F1 requiring correct boundaries and label.	Requires gold token/span annotations; none were evaluated here.

Appendix C — Recommended Next-Stage Work Plan

Priority	Next action	Completion evidence
1 — Data	Freeze a permitted, versioned dataset and publish a data card.	Source, licence, date, fields, counts, language profile.
2 — Annotation	Create category, urgency, entity and duplicate-pair annotation guidelines.	Guidelines and inter-annotator review.
3 — Classification	Run cross-validation and detailed error analysis.	Macro/per-class metrics and confusion matrix.
4 — NER	Annotate spans and train/evaluate one baseline token classifier.	Exact-span precision, recall and F1.
5 — Retrieval	Evaluate multilingual sentence embeddings on labelled pairs.	Precision@k / Recall@k and reviewer feedback.
6 — Integration	Build a controlled reviewer interface with evidence and abstention.	Functional and usability test results.
7 — Governance	Implement access, retention, audit and deletion controls.	Security/privacy checklist and test evidence.

This work plan is a recommendation for future development, not a statement that these tasks were completed during the six-week internship.

Appendix D — Glossary

Term	Meaning
NLP	Natural Language Processing: computational methods for analysing human language.
TF-IDF	Term Frequency–Inverse Document Frequency; a weighting scheme for text features.
LDA	Latent Dirichlet Allocation; a probabilistic topic-modelling method.
NER	Named Entity Recognition; identifying text spans and assigning entity labels.
BIO tagging	Sequence labelling convention using Beginning, Inside and Outside tags.
Tokenisation	Splitting text into tokens such as words or subword units.
Cosine similarity	A measure of similarity between vectors based on their angle.
Train-test split	Partitioning data for model fitting and held-out evaluation.
Data leakage	Use of information from outside the training data in a way that can inflate evaluation.
Human-in-the-loop	A workflow in which people review or correct model suggestions.
Code-mixed text	Text that combines languages or scripts within the same message.
Abstention	The system declines to provide a confident prediction and requests review.

Final evidence reminder

All reported numerical results in this document are transcribed from the weekly implementation reports and should be interpreted within their original experiment settings. Proposed targets, illustrative examples and future recommendations are not experimental findings.